Exam Date & Time: 01-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V Semester End Semester Make Up Examination Dec/Jan 25

BIOMATERIALS [BME 3123]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

1A)	What is the stress-shielding effect? What strategies could potentially address the issue.	(4)
1B)	Systematically analyze the strategies to harden/strengthen polymers.	(4)
1C)	Patients receiving mechanical heart valves often suffer with thromboembolism. How is this problem countered?	(2)
2A)	Compare cemented and cement less fixation process.	(3)
2B)	Compare bioprosthetic and mechanical heart valves.	(2)
2C)	Analyze the steps involved for the fixation of total hip arthroplasty.	(5)
3A)	Justify with reason why mitral and aortic valves are more susceptible to damage? Highlight the basic requirements for heart valve prosthesis.	(5)
3B)	Classify hydrogel contact lenses.	(2)
3C)	Illustrate the common desirable properties for contact lenses. Analyze how to increase oxygen permeability of hard contact lenses.	(3)
4A)	Classify polymer based on tacticity.	(3)
4B)	With the help of a diagram compare subperiosteal and endosteal dental implants. When are bone grafting and sinus lift performed? Justify with proper reason.	(5)

4C)	Explain the significance of healing screw during fixation of dental implant.	(2)
5A)	How do you convert a collagen-based hydrogel into a porous scaffold? Justify.	(2)
5B)	What are the roles of proline and hydroxyproline in the formation of collagen fibril? Susil intends to extract soluble collagen for the fabrication of some carrier matrices. Describe the steps involved in brief. Why is the mechanical strength in skin lower than tendon? Justify.	(5)
5C)	Shoham intends to make gelatin based porous scaffold. Propose a strategy to assist Shoham to fabricate porous gelatin scaffold (briefly, elaborate the process involved).	(3)

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